

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("student-mat.csv", sep=";")

df.head()
```

```
Out[1]:
```

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	fan
0	GP	F	18	U	GT3	A	4	4	at_home	teacher	...	
1	GP	F	17	U	GT3	T	1	1	at_home	other	...	
2	GP	F	15	U	LE3	T	1	1	at_home	other	...	
3	GP	F	15	U	GT3	T	4	2	health	services	...	
4	GP	F	16	U	GT3	T	3	3	other	other	...	

5 rows × 33 columns



```
In [2]: print("Dataset Shape:")
print(df.shape)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Rows:")
print(df.duplicated().sum())

print("\nData Types:")
print(df.dtypes)
```

Dataset Shape:
(395, 33)

Missing Values:

school	0
sex	0
age	0
address	0
famsize	0
Pstatus	0
Medu	0
Fedu	0
Mjob	0
Fjob	0
reason	0
guardian	0
traveltime	0
studytime	0
failures	0
schoolsup	0
famsup	0
paid	0
activities	0
nursery	0
higher	0
internet	0
romantic	0
famrel	0
freetime	0
goout	0
Dalc	0
Walc	0
health	0
absences	0
G1	0
G2	0
G3	0

dtype: int64

Duplicate Rows:
0

Data Types:

school	str
sex	str
age	int64
address	str
famsize	str
Pstatus	str
Medu	int64
Fedu	int64
Mjob	str
Fjob	str
reason	str
guardian	str
traveltime	int64
studytime	int64
failures	int64
schoolsup	str
famsup	str

```
paid          str
activities    str
nursery       str
higher        str
internet      str
romantic      str
famrel        int64
freetime      int64
goout         int64
Dalc          int64
Walc          int64
health        int64
absences      int64
G1            int64
G2            int64
G3            int64
dtype: object
```

```
In [3]: # Average Final Grade
print("Average Final Grade (G3):")
print(df["G3"].mean())

# Students scoring above 15
print("\nStudents scoring above 15:")
print((df["G3"] > 15).sum())

# Correlation
print("\nCorrelation between study time and final grade:")
print(df["studytime"].corr(df["G3"]))

# Gender Performance
print("\nAverage score by gender:")
print(df.groupby("sex")["G3"].mean())
```

```
Average Final Grade (G3):
10.415189873417722
```

```
Students scoring above 15:
40
```

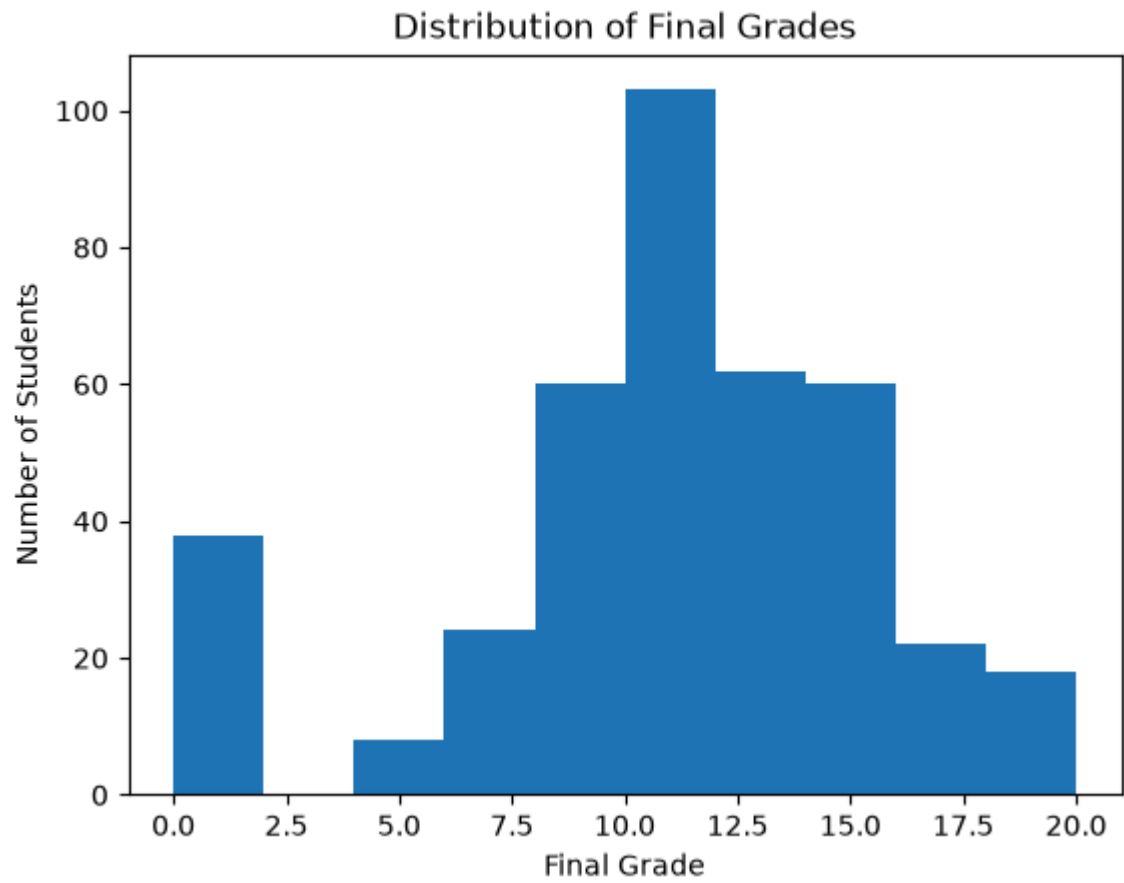
```
Correlation between study time and final grade:
0.09781968965319626
```

```
Average score by gender:
sex
F    9.966346
M   10.914439
Name: G3, dtype: float64
```

```
In [4]: plt.hist(df["G3"], bins=10)

plt.title("Distribution of Final Grades")
plt.xlabel("Final Grade")
plt.ylabel("Number of Students")

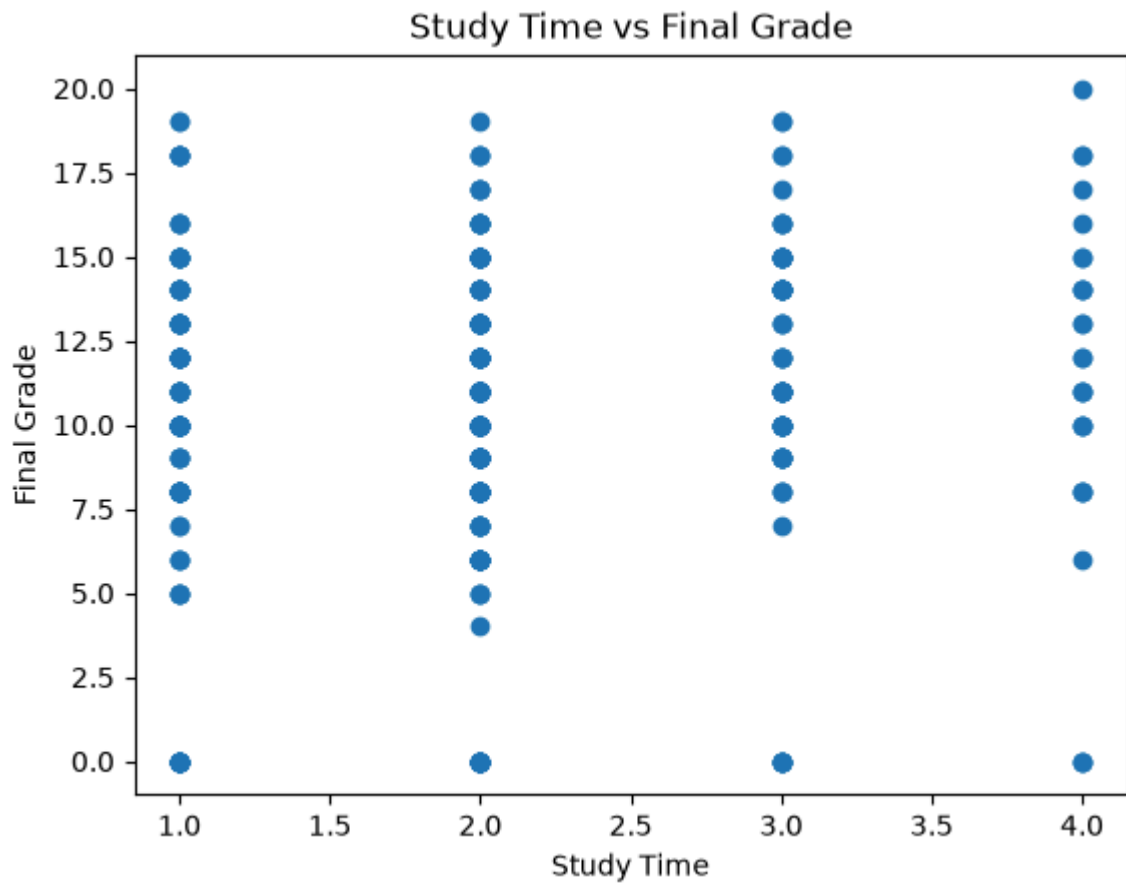
plt.show()
```



```
In [5]: plt.scatter(df["studytime"], df["G3"])

plt.title("Study Time vs Final Grade")
plt.xlabel("Study Time")
plt.ylabel("Final Grade")

plt.show()
```

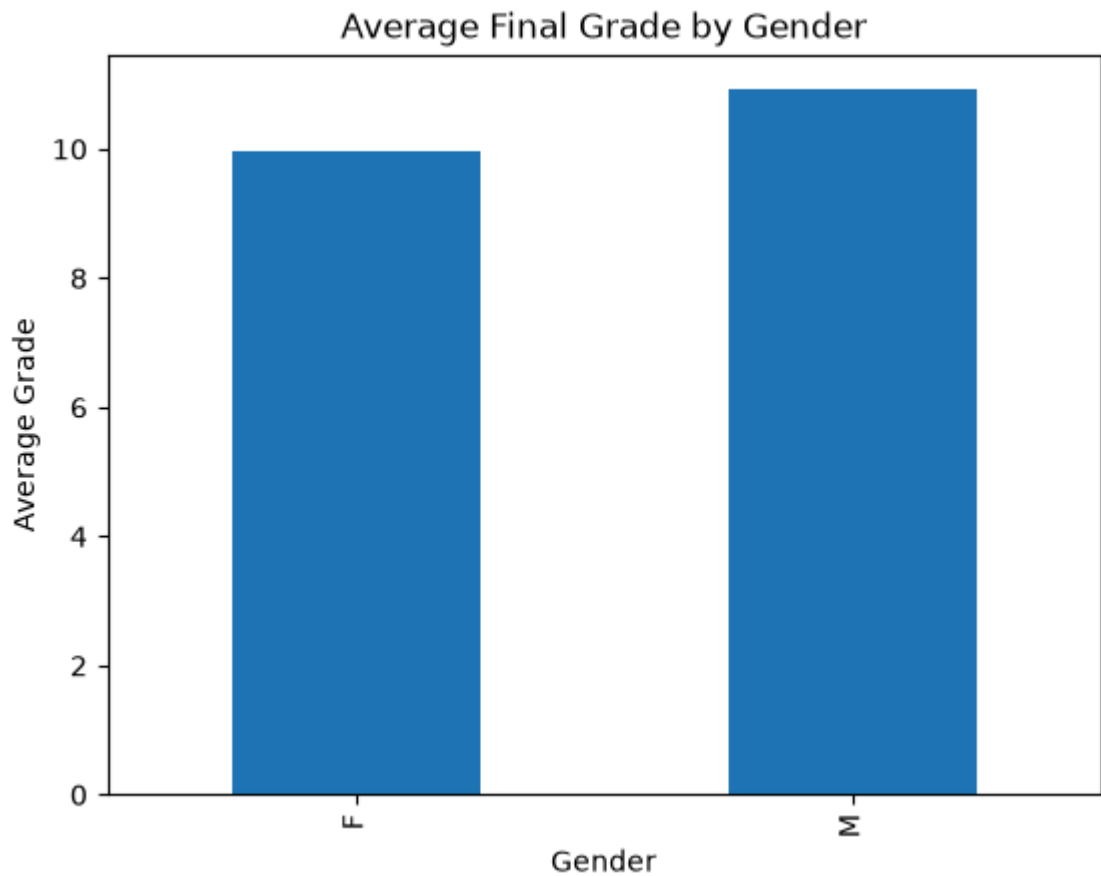


```
In [6]: gender_avg = df.groupby("sex")["G3"].mean()

gender_avg.plot(kind="bar")

plt.title("Average Final Grade by Gender")
plt.xlabel("Gender")
plt.ylabel("Average Grade")

plt.show()
```



Conclusion

The dataset contains 395 students and 33 variables.

No missing values or duplicate records were found.

The average final grade (G3) is approximately 10.42.

40 students scored above 15.

Study time shows a weak positive relationship with final grades.

Male students performed slightly better on average than female students.

Visualizations helped understand grade distribution, study habits, and gender-wise performance.

In []: